

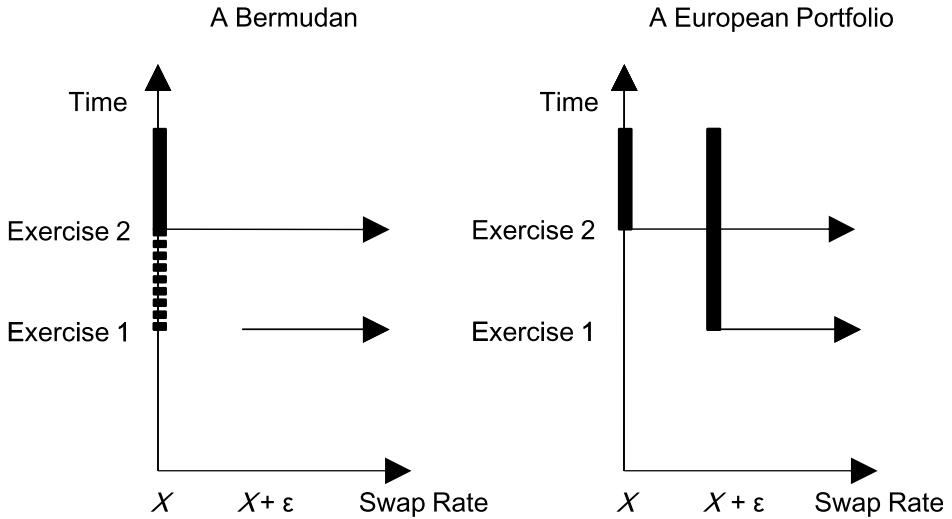


Figure 5.19: A Bermudan Payer Swaption Struck at X with Two Exercise Dates and a Replicating Portfolio of a Short-Dated European Payer Swaption Struck at $X + \epsilon$ and a Long-Dated European Payer Struck at X

Source: Zhou (2003).

cised by the buyer at either Exercise 1 or Exercise 2. This Bermudan payer is represented in the left-hand panel of Figure 5.19. The underlying swap is represented by the thick line along the time axis. Due to the two exercise times, it may be the case that exercise is not elected until the second exercise date (not to mention the fact that exercise may not occur at all), in which case the front portion of the underlying swap would not be realized upon exercise. This is denoted by the dashed portion of the thick line of the underlying swap.

Each of the two horizontal lines with arrows indicates the range of swap rates for which the Bermudan option would be exercised on its respective exercise dates. Although the strike of this Bermudan payer swaption is X , it is important to note that if the underlying swap rate at the time of the first exercise is just slightly in-the-money, the holder of the option may rationally choose to forgo exercise at that point in time. If the value of the remaining option that expires at the second exercise date is greater than the value associated with exercising the option at the first exercise date, then the option will not be exercised on the first exercise date.⁴⁶

⁴⁶We saw a situation like this in Example 1 of Section 5.6.1.

However, at the time of the first exercise date, there is some swap rate, denoted $X + \epsilon$, at which it will make sense to exercise the swaption. The Bermudan will be optimally exercised, provided that the then-prevailing swap rate on the underlying swap is at or above $X + \epsilon$. This optimal exercise region is denoted by the horizontal arrow at Exercise 1 that extends to the right starting at $X + \epsilon$.⁴⁷ If the second exercise time is reached, the Bermudan option will be exercised anytime the then-prevailing swap rate on the remaining underlying swap is above the strike X , since there is no further opportunity for exercise and the option is in-the-money. This exercise range is denoted by the second horizontal line at Exercise 2.

The payoff profile of the Bermudan can (more or less) be replicated by a portfolio of two European payer swaptions, represented in the right panel of Figure 5.19. The two underlying swaps of the two European payers are represented, respectively, by the two thick vertical lines. One of these is a long line, representing the exercise of the short-dated European payer expiring at the first exercise date and struck at $X + \epsilon$ into the same underlying swap corresponding with exercise of the Bermudan at its first exercise date. The shorter of the two vertical lines represents the exercise of the long-dated European payer expiring at the second exercise date and struck at X into the shorter swap, corresponding with exercise of the Bermudan at its second exercise date. The two horizontal lines with arrows represent ranges in which each of the Europeans would each be exercised.

Consider now replicating \$100mm of the Bermudan payer struck at X by

1. Buying \$100mm of the long-dated European payer struck at X and
2. Buying \$Nmm of the short-dated European payer struck at $X + \epsilon$,
3. Selling back into the market the long-dated payer at its then-prevailing price at the time of the first exercise if the short-dated European payer happens to be exercised.

Notice that the Bermudan and the portfolio of Europeans will be exercised in the same states of the world and into the same underlying swaps. At the first exercise date the Bermudan will be exercised if the then-prevailing swap rate on the longer underlying swap is above $X + \epsilon$. The short-dated European will be exercised in this case as well. At the second exercise date the Bermudan will be exercised (assuming there was no exercise at the first date) if the then-prevailing swap rate on the shorter underlying swap is above

⁴⁷This is not to suggest that the threshold swap rate $X + \epsilon$ can be determined today with certainty. As we saw in Examples 2 and 3 of Section 5.6.1, the decision to exercise depends not only on the level of rates but also on the level of implied vol for the residual swaption prevailing at the time of first exercise.

X . The long-dated European will be exercised in this case as well (assuming there was no exercise of the short-dated European, which would have meant that the long-dated European would have been sold back to the market in accordance with the replicating strategy outlined above).

The notional on the long-dated European must be the same as the Bermudan so that the two have identical cash flows if exercise occurs at the second exercise date. However, the notional on the short-dated European will in general be different from the \$100mm notional on the Bermudan. If the short-dated European is exercised at the first exercise date (which means that the Bermudan has also been exercised on the first exercise date), the long-dated European will be sold back into the market at its then-prevailing price. The proceeds from this sale can be used to lower the notional required on the short-dated European to match the value generated by the Bermudan.⁴⁸

⁴⁸It is important to note that although this heuristic replication shows the relationship between Bermudans and a portfolio of Europeans of varying expiries, strikes, and underlyings, there are a number of nuances that we have (intentionally) glossed over. First, the threshold swap rate $X + \epsilon$ above which the Bermudan should be exercised on the first exercise date is determined on the first exercise date itself. How, then, can we consider buying today the shorter-dated European struck at $X + \epsilon$ if that strike rate is not yet known? Second, we indicated that as part of the replication strategy the longer-dated European will be sold on the first exercise date back into the market at its then-prevailing price if the shorter-dated European is exercised at the first exercise date. The proceeds of this sale can be used to lower the amount of notional on the short-dated European needed to replicate the value of the Bermudan in the event of exercise at the first exercise date. Since the proceeds from this sale will not be known until the first exercise date, how do we know by how much we can reduce the notional of the short-dated European that we are buying today? We don't know the answer to this question with certainty today. We will only know the price at which we can sell the long-dated European at the time of the first exercise date once we get to the first exercise date, unless of course we could lock in that price somehow today. The then-prevailing price will depend on the swap curve prevailing at the time, implied vol prevailing at the time, and skew prevailing at the time. Nonetheless, we can conceptualize locking in some forward price for this swaption today. This alludes to a concept known as *forward vol*, which we will discuss in further depth in Chapter 9. The link to forward vol is one of the complicating issues in the analysis of Bermudan swaptions.

Finally, notice that if exercised at the first exercise date, the Bermudan will be worth the value of an annuity equal to $R - X$ paid over the life of the full underlying swap, where $R > X + \epsilon$ is the then-prevailing swap rate for the longer underlying swap, whereas the short-dated European (which will be exercised as well) will be worth the value of an annuity equal to $R - (X + \epsilon)$ paid over the life of the full underlying swap. The value of this annuity is less than that paid by the Bermudan. This effect will tend to raise the notional required on the shorter-expiry European. We could remedy this latter shortfall by incorporating a European binary option struck at $X + \epsilon$ that expires at the first expiration date and pays an annuity of ϵ over the life of the long underlying swap if the underlying swap rate at expiry is above $X + \epsilon$, but that would seem to unnecessarily complicate our heuristic analysis.

One final important point to consider is that the analysis of Bermudans is inexorably linked to a consideration of skew. To see this, let us consider the very simplistic case in which the swap curve is perfectly flat and all swap rates are equal to a rate we will denote R_0 . In this case, all forward swap rates obtained through bootstrapping will also be equal to R_0 . Now consider the portfolio of two European payers in the replication of the Bermudan. The long-dated payer is struck at X and the short-dated payer is struck at $X + \epsilon$, and thus at least one of these swaptions is not struck at-the-money forward. Thus, skew must be considered for at least one of these two options. For any arbitrary swap curve, in general both strikes in the European portfolio will be away from the money, and skew must be considered when we value both of the options.

Problem Set 5

1. Consider again our stylized example in which swaps pay annual bond on the fixed side versus 1-year LIBOR. A 5x10 floor on 1-year LIBOR struck at 4% has its first reset date five years from now, first payment date six years from now and final payment date ten years from now. Consider now a 5 into 5 European style receiver swaption struck at 4% annual bond versus 1-year LIBOR. The owner of this option has the right to receive 4% fixed in a swap that is effective five years from today and has a final maturity that is ten years from today.

Which is worth more, the 5x10 4% floor or the 5 into 5 4% receiver swaption? Explain.

2. Consider our stylized example in which swaps are quoted annual bond on the fixed side versus 1-year LIBOR quoted annual bond as well. Suppose the swap curve is as follows:

Maturity	Swap Rate
1 year	5.00%
2 year	5.25%
3 year	5.50%
4 year	5.75%
5 year	6.00%

Suppose that we use Black's model to price up swaptions and that the lognormal swaption matrix prevailing at this time is as given in Table 5.1. Further suppose that there is no skew. Suppose that a client buys \$100mm of a 2 into 3 swaption straddle struck at-the-money forward from a dealer.

a) What is the price that the client pays today for the straddle?

b) Suppose that the client holds the position for a year and that after a year swap rates are unchanged and the lognormal swaption matrix is unchanged. That is, after a year swap rates are as given above and swaption vols are as they appear in Table 5.1. What is the value of the straddle after one year?

3. Can you conclusively answer the following questions:

a) Which is worth more: a 1 into 4 7% receiver or a 2 into 3 7% receiver?

b) Which is worth more: a 1 into 4 7% payer or a 2 into 3 7% payer?

In each case, explain why one is always worth more than the other or provide specific pricing examples to show that the price of one is not always greater than the price of the other.

4. Does swaption parity hold under the normal model? That is, if swaption prices are determined by the normal model, does swaption parity hold?

5. Suppose that the swap curve is as follows:

Maturity	Swap Rate
1 year	6.00%
2 year	6.25%
3 year	6.50%
4 year	6.75%
5 year	7.00%

Assume that these rates are annual bond coupons versus 1-year LIBOR annual bond as well. Assume the notional on all trades is \$100mm. Unless stated otherwise, all pricing is done at mid-market. Use Black's formula to price swaptions. Use the lognormal swaption vol matrix in Table 5.1 to determine the appropriate mid-market volatility and assume flat skew.

a) What is the value today of a 2 into 3 payer swaption, struck at 7%?

b) What is the value today of a 2 into 3 receiver swaption, struck at 7%?

c) Use swaption parity to determine the value of a 2x5 forward starting swap with a coupon of 7%. Verify the value of the swap by pricing up this forward starting swap.

Use the following swap curve for questions 6–7:

Maturity	Swap Rate
1 year	5.00%
2 year	5.25%
3 year	5.50%
4 year	5.75%
5 year	6.00%

Assume that these rates are annual bond coupons versus 1-year LIBOR annual bond as well.

6. For this question, use the normal model to price swaptions. Assume the notional on all trades is \$50mm. Unless stated otherwise, all pricing is done at mid-market. Use the normal swaption vol matrix in Table 5.3 to determine the appropriate mid-market volatility and assume *flat normal skew* (meaning that the level of normal implied vol is assumed to remain constant as we move the strike of the swaption).

- a) What is the value today of a 1 into 3 payer swaption, struck at 7%?
- b) Given the level of normal implied vol that you used to price up the swaption in part a), what is the corresponding level of lognormal implied vol in the lognormal model?
- c) What is the value today of a 1 into 3 receiver swaption, struck at 5%, assuming flat normal skew?
- d) Given the level of normal implied vol that you used to price up the swaption in part c), what is the corresponding level of lognormal implied vol in the lognormal model?
- e) How do the two levels of lognormal implied volatilities that you found in part b) and part d) compare? Explain.

7. The purpose of this question is to derive (numerically) the relationship between σ_N in the normal model and σ_{LN} in the lognormal model under the assumption that normal implied vol is constant regardless of the strike of the option. That is, we wish to derive a relationship similar to the one in Figure 5.8.

To this end, consider our stylized example and 1 into 4 payer swaptions of various strikes. Assume that 1 into 4 payer swaptions are priced with a con-

stant normal implied vol, $\sigma_N = 124$ bps, that does not vary as a function of strike. We will use the procedure outlined in the last section of Appendix E to determine the relationship between σ_N and σ_{LN} .

a) In particular, consider the following nine strikes: ATMF $- 200$ bps; ATMF $- 150$ bps; ATMF $- 100$ bps; ATMF $- 50$ bps; ATMF; ATMF $+ 50$ bps; ATMF $+ 100$ bps; ATMF $+ 150$ bps; ATMF $+ 200$ bps. Compute the price for each of these nine swaptions under the normal model using $\sigma_N = 124$ bps; and in each case solve for the level of σ_{LN} in the lognormal model that equates to the same premium.

b) What is the level of σ_{LN} that you solve for when considering the ATMF strike?

c) How does this level of σ_{LN} that you solve for in part b) compare with the approximation obtained using Equation 5.27?

d) Provide a graph in which you allow strikes to vary along the x-axis and the level of lognormal implied vol (in percentage terms) to vary along the y-axis. In particular, show two curves in this graph:

- (1) A flat line equal to the level of σ_{LN} you solve for in part b);
- (2) A curve that shows how σ_{LN} must vary by strike under the assumption of constant normal implied vol for the nine strikes you have considered.

Interpret your graph.

8. For this question, use the lognormal model to price swaptions and assume flat skew.

a) Consider our stylized example and suppose we are long \$100mm of a 1 in 2 Bermudan receiver swaption struck at 4.75% with annual exercise (i.e., this swaption may be exercised in one year or in two years). Suppose that after one year the swap curve is as follows:

Maturity	Swap Rate
1 year	4.00%
2 year	4.25%
3 year	4.50%
4 year	4.75%
5 year	5.00%

These rates are annual bond versus 1-year LIBOR. Suppose further that the lognormal swaption matrix prevailing at this time is as given in Table 5.1.

Should we exercise our receiver swaption at this time?

b) Consider again our stylized example and suppose we are long a particular type of Bermudan swaption. This is a 1 into 4 Bermudan receiver; however, it may only be exercised one year from now and two years from now. If it is exercised in one year, the owner of the option will receive fixed in a 4-year swap. If it is not exercised in a year, the owner of the option has the right to exercise one year thereafter (i.e., two years from today) and upon doing so will receive fixed in a 3-year swap. These are the only two times that this option may be exercised. The strike on the swaption is 5.25%.

Suppose that in one year the swap curve is as given in part a) of this question and that the swaption matrix prevailing at this time is as in Table 5.1. Should we exercise our option in a year?

9. Denote by $p(X)$ and $r(X)$ the price of a payer swaption and receiver swaption struck at X , respectively. Denote by F the forward swap rate on the underlying swap in these swaptions. Out-of-the-money payer swaptions and receiver swaptions are *symmetric in price about the forward* if

$$r(F - \epsilon) = p(F + \epsilon)$$

for every $\epsilon > 0$.

a) Are out-of-the-money payer swaptions and receiver swaptions symmetric in price about the forward under the normal model? If so, prove it. If not, find a counterexample to show that this is not true.

b) Are out-of-the-money payer swaptions and receiver swaptions symmetric in price about the forward under the lognormal model? If so, prove it. If not, find a counterexample to show that this is not true.

10. A *swaption straddle* is a position consisting of both a receiver swaption and a payer swaption, both with the same expiry, underlying, and strike. A swaption straddle is struck at-the-money forward if the common strike on the payer and receiver is the at-the-money forward strike for the payer and receiver. Suppose that a client sells an at-the-money forward swaption straddle to a dealer. Assume the trade is held to expiration by both counterparties and that the underlying swap rate finishes *exactly* at-the-money.

Is it possible that both counterparties can have bona fide positive profits from this trade? If so, explain how so. If not, explain why not.

11. Consider our stylized example in which swaps are quoted annual bond on the fixed side versus 1-year LIBOR quoted annual bond as well. Suppose the swap curve is as follows:

Maturity	Swap Rate
1 year	4.00%
2 year	4.25%
3 year	4.50%
4 year	4.75%
5 year	5.00%

Suppose that we use Black's model to price up swaptions and that the log-normal swaption matrix prevailing at this time is as given in Table 5.1. Further suppose that there is no skew. Consider the following trade between a dealer and a client:

1. The dealer will pay 1-year LIBOR + 75 bps to the client for four years
2. The client will pay 1-year LIBOR + 50 bps to the dealer for four years
3. The dealer has the one-time right in one year to convert the cash flows paid by the client to a fixed rate for the remaining three years of the trade. If this right is not exercised by the dealer, then the client will continue to pay 1-year LIBOR + 50 bps for the remaining three years of the trade.

Assume that this trade has a notional of \$100mm. What fixed rate should be quoted in this trade to make it an NPV = 0 transaction?

12. This question is designed to show that in addition to pricing differences that may arise between the normal model and the lognormal model, the way dealers hedge themselves can depend very much upon which model they employ. You may find it useful to utilize the results given in Section E.3 of Appendix E.